Title: NYAYaRAG: A Retrieval-Augmented Generation Framework for Legal Question Answering over Indian Supreme Court Judgments

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Repository: https://github.com/harshalichowdhary-design/NyayaRAG

1. Introduction

1.1 Background and Context

The Indian judiciary operates under conditions of extreme structural strain: more than 23,000 judges across all tiers contend with a persistent, well-documented case backlog (Nigam et al., 2023). Article 141 of the Constitution of India compounds the stakes of this strain by making Supreme Court judgments binding law on every court in the country, so efficient, accurate retrieval of relevant precedent is a functional requirement of the legal system, not a convenience. Legal research practice has historically relied on keyword and Boolean search, which matches surface-level tokens rather than legal meaning and performs poorly when a query and a relevant judgment use different vocabulary for the same concept (Martinez-Gil, 2023); legal NLP has since progressed toward Transformer-based models and, most recently, retrieval-augmented generation (RAG) architectures pairing a retrieval component with a generative model (Hartung et al., 2023).

RAG, as formalized by Lewis et al. (2020), grounds a generative model's output in retrieved evidence rather than parametric knowledge alone, reducing factual drift — especially consequential in law, where an answer's value depends on whether it is traceable to an authoritative source. Grounding is not a guarantee against error: a model conditioned on irrelevant or misapplied evidence can still produce a fluent but legally incorrect answer (hallucination), and recent surveys argue reliability and explainability — not fluency — should determine whether legal AI can be trusted (Nguyen et al., 2025). Little existing work directly evaluates retrieval-strategy choices — lexical, dense, hybrid — for Indian Supreme Court legal QA, or connects that comparison to a generation model's behavior under both answerable and deliberately unanswerable conditions; NyayaRAG is motivated directly by this gap.

1.2 Problem Statement

There is limited empirical evidence, specific to Indian Supreme Court legal question answering, on how retrieval strategy affects retrieval effectiveness, downstream answer quality, and abstention reliability when evidence is insufficient. Prior work establishes that semantic retrieval can outperform lexical retrieval in general and adjacent legal settings (Nigam et al., 2023; Karpukhin et al., 2020), but does not provide a controlled comparison across multiple lexical, dense, and hybrid configurations on a single expert-constructed Indian Supreme Court benchmark, nor connect retrieval performance to grounded-answer quality and safe abstention under matched conditions. This study addresses that gap through a controlled comparison of six retrieval configurations — TF-IDF, BM25, E5, BGE, InLegalBERT, and a

Hybrid BM25–E5 model combined via Reciprocal Rank Fusion — against retrieval effectiveness, generated-answer quality on answerable questions, and abstention accuracy on deliberately unanswerable questions, with all other factors (corpus, chunking, benchmark, retrieval depth k = 5, generation temperature = 0.0) held constant to isolate retrieval strategy as the manipulated variable.

1.3 Purpose of the Study

The purpose of this study is to design, implement, and empirically evaluate NyayaRAG in order to determine which retrieval strategies most effectively surface relevant precedent and how retrieval quality translates into evidence-grounded, trustworthy generated answers — producing statistically validated evidence, rather than descriptive comparison, and assessing whether the resulting system can generate accurate answers while reliably abstaining when evidence is insufficient. In doing so, the study contributes both a reusable evaluation methodology for legal RAG systems and a working, benchmarked prototype relevant to the Indian judicial context.

1.4 Research Questions

- **RQ1:** Do semantic retrieval models significantly outperform lexical retrieval models for Indian Supreme Court legal question answering?
- **RQ2:** Which single retrieval model performs best overall among TF-IDF, BM25, E5, BGE, InLegalBERT, and the Hybrid BM25–E5 retriever?
- **RQ3:** How accurately does NyayaRAG generate evidence-grounded answers?
- **RQ4:** Can NyayaRAG reliably detect insufficient evidence and abstain?

1.5 Contributions and Expected Value

NyayaRAG contributes at both a practical and a technical level. Practically, given a legal question, the system retrieves relevant precedent, generates an answer that cites its supporting chunks, and — critically — is designed to recognize when the evidence does not support a confident answer and to abstain rather than generate an unsupported response; traceability of every answer back to specific retrieved text is a core design requirement, addressing the trustworthiness concerns raised in the legal AI literature (Nguyen et al., 2025; Magesh et al., 2024). Technically, the study contributes a controlled, six-configuration comparison of retrieval strategies, an expert-generated legal benchmark built from questions and relevance judgments contributed by practicing judicial officers rather than crowd-sourced or synthetic queries, and a statistically rigorous evaluation protocol (Wilcoxon signed-rank, Friedman with Holm-corrected post-hoc tests, exact binomial tests).

The expected value of this work accrues primarily to judicial officers, advocates, and legal researchers — the professional user base the benchmark was built around. This study does not evaluate the system with citizen or lay-user participants, and no claims are made regarding its suitability for direct citizen-facing legal advice; any extension in that direction would require separate, dedicated evaluation.

2. Literature Review

2.1 Literature Review Approach

The literature review draws on ACL Anthology, arXiv, IEEE Xplore, and Google Scholar, supplemented by targeted searches of legal-technology and information-retrieval venues, combining legal-domain keywords ("legal NLP," "Indian judiciary," "case law retrieval," "Article 141") with technical keywords